

Attitude inversion of space debris based on the laboratory-tested photometry dataset

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Abstract

The attitude of space debris holds significant implications for active debris removal and collision warnings. This study constructs a comprehensive photometry dataset for the target, utilizing angle transformation relationships and drawing upon the CZ-4C photometric data tested in the laboratory. The photometry in the dataset with noise and translation simulates the telescope observed one. We use the genetic algorithm to search for attitude parameters of the photometry that matches the simulated photometry in the dataset under the constraint of attitude angle rules. The derived and preset values of attitude parameters exhibit a strong alignment, applicable to light curves with an amplitude of about 2.5 magnitudes and noise levels up to 1.0 magnitudes. This suggests that the algorithm's accuracy is validated against the preset values, demonstrating its robust resistance to noise. Furthermore, the algorithm proposed in this study for deriving attitude from photometry requires a small amount of photometry data. This study provides a promising algorithm for future attitude analyses from telescope-observed photometry.

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1. Introduction

With the advancement of human space activities, the amount of space debris generated by spacecraft accidents and satellite retirements continually increases. The debris not only occupies limited orbital resources but also threatens the safety of on-orbit spacecraft. Developments of active debris removal (ADR) technology and space traffic management strategy are crucial to protect space assets and ensure sustainable of space endeavors. Current ADR methods include electrodynamic tether, laser ablation, and electro-orbit raise tether, all of which tailor their strategies based on the attitude of space debris to enhance

removal efficiency and success rates. Besides, attitude alterations in space debris can affect the aerodynamic drag it experiences, subsequently leading to perturbations and drift that deviate from predictions. Accurate attitude information allows scientists to more precisely calculate the trajectory of space debris. Thereby it can enhance the accuracy of future position predictions and better assessing the collision probability with other orbital objects, such as satellites or spacecraft, to implement appropriate collision avoidance measures. Consequently, research into the attitude of space debris is of significant importance for devising more scientifically sound space traffic management strategies (McKnight, 2019; Skinner et al., 2019; Piergentili et al., 2022; International Institute for Space Law, 2022), ensuring the safety and fluidity of space traffic.

In recent years, researchers have made substantial progress in studying space debris attitude. Their methods can

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be divided into three categories broadly. The first category is refining the target's force model with the analysis of target properties and environmental conditions. It involves the eddy current torque from the metallic parts (Pritykin et al., 2018), the aerodynamic drag experienced by solar panels (Gao et al., 2022), and the YORP effect generated by thermal radiation (Lin, 2022; Cuomo, 2023). Then they perform a long-term evolution analysis of target attitude angles under refined force models. The second category focuses on the trend of photometry over time, and studies the periodicity of the light curves to analyze the rules of target attitude angles (Binz et al., 2014), thereby achieving the inversion of the spin axis direction. The third category specifically addresses targets equipped with solar panels (Song et al., 2020; Payne et al., 2006; Hall, 2011; Hinks and Crassidis, 2016). Researchers utilize the peak values in the light curve to ascertain the target's rotational period and operational status. These aforementioned methods have achieved partial attitude information of some targets, inspiring confidence in inverting the space debris attitude from light curves. However, these methods all require long-term photometry observation, making it challenging to broadly apply them to the general analysis of space debris attitude. Even when long-term photometry is available, if the periodicity or noise at different times is unstable, or if the peak photometric values have not been obtained due to the constraints such as observation angle, the inversion accuracy of these methods is reduced.

In response, some researchers have attempted to study the inversion methods that do not rely on long-term observation data. The Optical Test Center of the National Aeronautics and Space Administration (NASA) established a ground laboratory to get images of the SL-8 Cosmos 3 M second-stage rocket body under different attitude angles and surface materials (Cowardin et al., 2010). The University of Rome used numerical simulation to investigate the photometry of Tiangong-1 under different attitudes (Piergentili et al., 2020). The Japan Aerospace Exploration Agency (JAXA) used laboratory test to preliminarily determine H2A R/B attitude type from telescope observed photometry (Yanagisawa et al., 2021). Then they analyze its detailed attitude angles by numerical simulation. Building upon these works, we use the simulated photometry dataset from laboratory test to inverse the attitude parameters of a single arc photometry. By correlating data from multiple arcs, the attitude solutions can be constrained to a relatively precise range. The inversion algorithm presented in this paper enhances the reliability and accuracy of the attitude while reducing the requirements for the quantity and characteristics of photometry.

This paper takes the CZ-4C rocket upper stage as the research object. In the optical observation simulation laboratory, the photometry of a 1/9 scale model of the CZ-4C rocket upper stage is tested. Through coordinate transformation, we construct a comprehensive photometry dataset, providing a data foundation for attitude inversion research. During the inversion, the attitude angles rule is

constrained by non-external torque. The attitude parameters are searched by the genetic algorithm, and the accuracy is verified against preset values.

2. Photometry dataset and attitude angles

Our methodology involves searching in the photometry dataset for the light curve that aligns with the simulated one. The inferred attitude parameters of the inversed photometry are then regarded as those of the simulated one. The tasks involves four key steps. First, we should ascertain the relationship between the target attitude angles in the laboratory and space. Next, we build the photometry dataset based on laboratory test. Thirdly, we explore the rules of the target attitude angles in the absence of external torque. Finally, we search for attitude parameters in the photometry dataset, and improve the accuracy of the attitude solution through the correlation of multi-arc data.

2.1. The relationship of attitude angles in the laboratory and space

We gather the photometric data on a 1/9 scale model of the CZ-4C rocket upper stage within our laboratory. The scale model meticulously replicates the geometric shape and material composition of the actual rocket prior to flight, with the dimensions of the model being approximately $0.3 \times 0.3 \times 0.6 \text{ m}^3$. The schematic diagram of the laboratory is shown in Fig. 1. The beam of light emitted from the source illuminates the target and is then received by the detection camera following reflection off the target. The spectral distribution of the light source is highly consistent with that of solar radiation, and the detection camera is the Andor 4040. In the laboratory, we change the target attitude angle by rotating the stand of the target. Additionally, we manipulated the observation phase angle β by rotating the camera stand, as detailed in our previous publication (Wang et al., 2023).

We need to introduce the coordinate systems utilized in this paper to define the target attitude angles in both the laboratory and space. In the laboratory, we designate the target's center of mass as the origin point, denoted as o . The direction in which the target points towards the light source is identified as the positive direction of the oX^o axis. The plane where the target, light source, and camera are located is defined as the oX^oY^o . These definitions form the laboratory center-of-mass-to-light-source coordinate system $oX^oY^oZ^o$, as depicted in Fig. 1. We construct the body-fixed coordinate system $oX^*Y^*Z^*$ by taking the rotation symmetric axis of the CZ-4C main body as the oZ^* axis, and the arrangement direction of the two tail adapters as the oY^* axis direction, as demonstrated in Fig. 2 (a). In the laboratory stand rotation system, the initial positions of rotation axes 1 and 3 coincide with the oZ^o direction, and the initial position of rotation axis 2 aligns with the oX^o direction. We then establish the correlation between

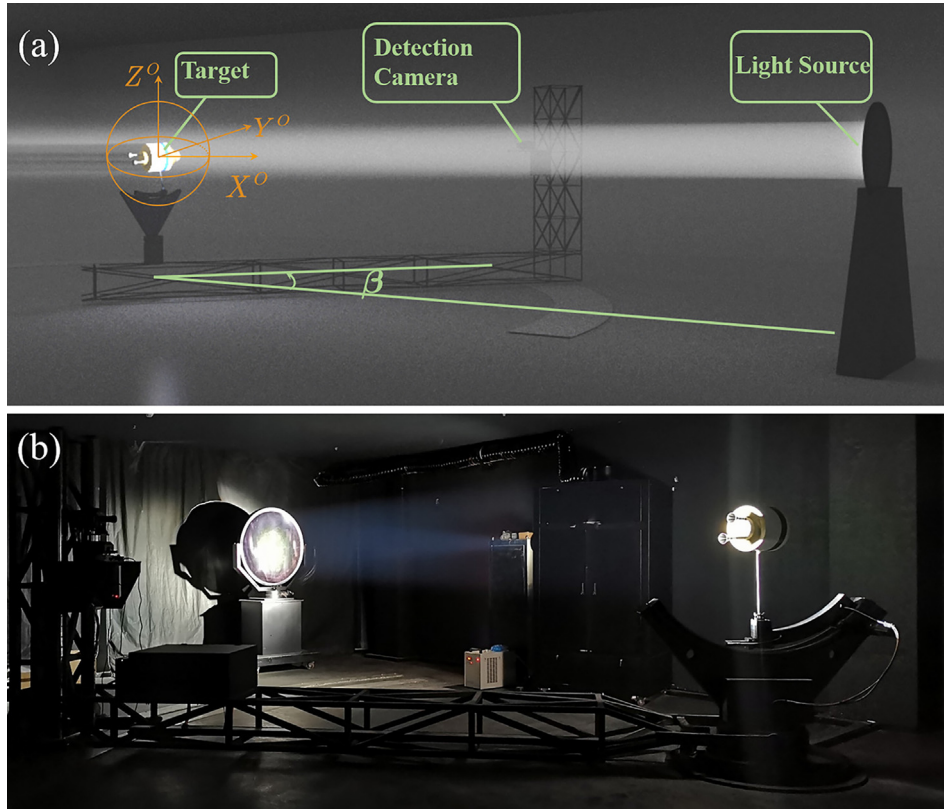


Fig. 1. Laboratory Setting. (a) Schematic diagram. (b) Actual image.

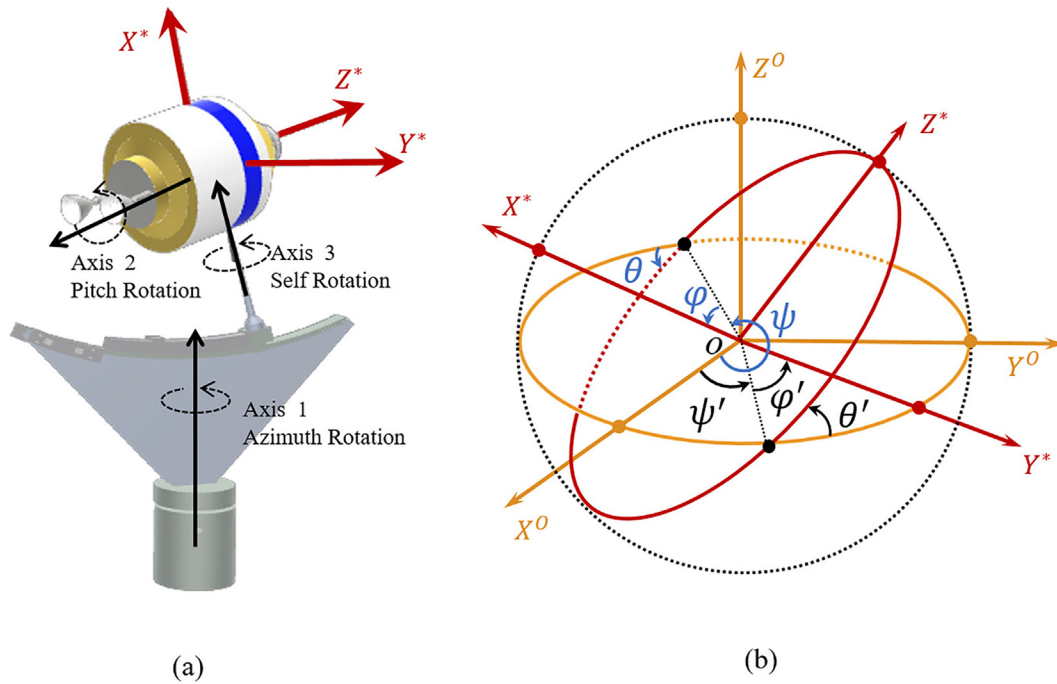


Fig. 2. Relationship of Laboratory Coordinate Systems. (a) Schematic diagram illustrating the relationship between the target body-fixed coordinate system $oX^*Y^*Z^*$ and the laboratory rotation axis system. (b) Schematic diagram of the rotational relationship from the target body-fixed coordinate system $oX^*Y^*Z^*$ to the center-of-mass-to-light-source coordinate system $oX^oY^oZ^o$.

the laboratory stand rotation angles (ψ', θ', ϕ') and the body-fixed coordinates $oX^*Y^*Z^*$ attitude angles (ψ, θ, ϕ)

in the center-of-mass-to-light-source coordinate system $oX^oY^oZ^o$. This relationship is as follows,

$$\begin{aligned} & \mathbf{R}_x(\varphi')\mathbf{R}_y(\theta')\mathbf{R}_x(\psi')\mathbf{R}_y(-90^\circ)\mathbf{R}_z(-90^\circ) \\ &= \mathbf{R}_z(\varphi)\mathbf{R}_x(\theta)\mathbf{R}_z(\psi), \end{aligned} \quad (1)$$

where the notation $\mathbf{R}_A(B)$ represents a rotation angle B around axis A .

$$\begin{aligned} \mathbf{R}_x(B) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos B & \sin B \\ 0 & -\sin B & \cos B \end{pmatrix}, \\ \mathbf{R}_y(B) &= \begin{pmatrix} \cos B & 0 & -\sin B \\ 0 & 1 & 0 \\ \sin B & 0 & \cos B \end{pmatrix}, \\ \mathbf{R}_z(B) &= \begin{pmatrix} \cos B & \sin B & 0 \\ -\sin B & \cos B & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned} \quad (2)$$

We establish the target's coordinate systems in the space, as shown in Fig. 3. The center-of-mass-to-light-source coordinate system, denoted as $oX^oY^oZ^o$, is aligned with that in the

laboratory. The plane containing the target, light source, and camera is designated as the oX^oY^o plane, with the target's center of mass serving as the origin, denoted by o . The oX^o axis is defined in the direction from o to the light source. We define the ecliptic coordinate system, denoted as $oX_SY_SZ_S$, with its origin at the target's center of mass. The oX_S axis is oriented towards the solar, while the oZ_S axis aligns with the direction of the ecliptic pole, as shown in Fig. 3 (a). The $oX^oY^oZ^o$ axis of the coordinate system $oX_SY_SZ_S$ is aligned with the oX_S axis of the ecliptic coordinate system $oX_SY_SZ_S$. Let α represent the angle between the oX^oY^o plane and the oX_SY_S plane. The rotation matrix transforming coordinates from the $oX^oY^oZ^o$ system to the $oX_SY_SZ_S$ system can then be expressed as $\mathbf{R}_x(-\alpha)$. The inertial coordinate system as the J2000.0 epoch equatorial coordinate system is denoted by $oX'Y'Z'$, with the target's center of mass as the origin. The rotation matrix for the transformation from the $oX_SY_SZ_S$ coordinate system to the $oX'Y'Z'$ coordinate system is $\mathbf{R}_x(-\varepsilon)\mathbf{R}_z(-(\omega + M))$, where ε represents the ecliptic equatorial angle, ω repre-

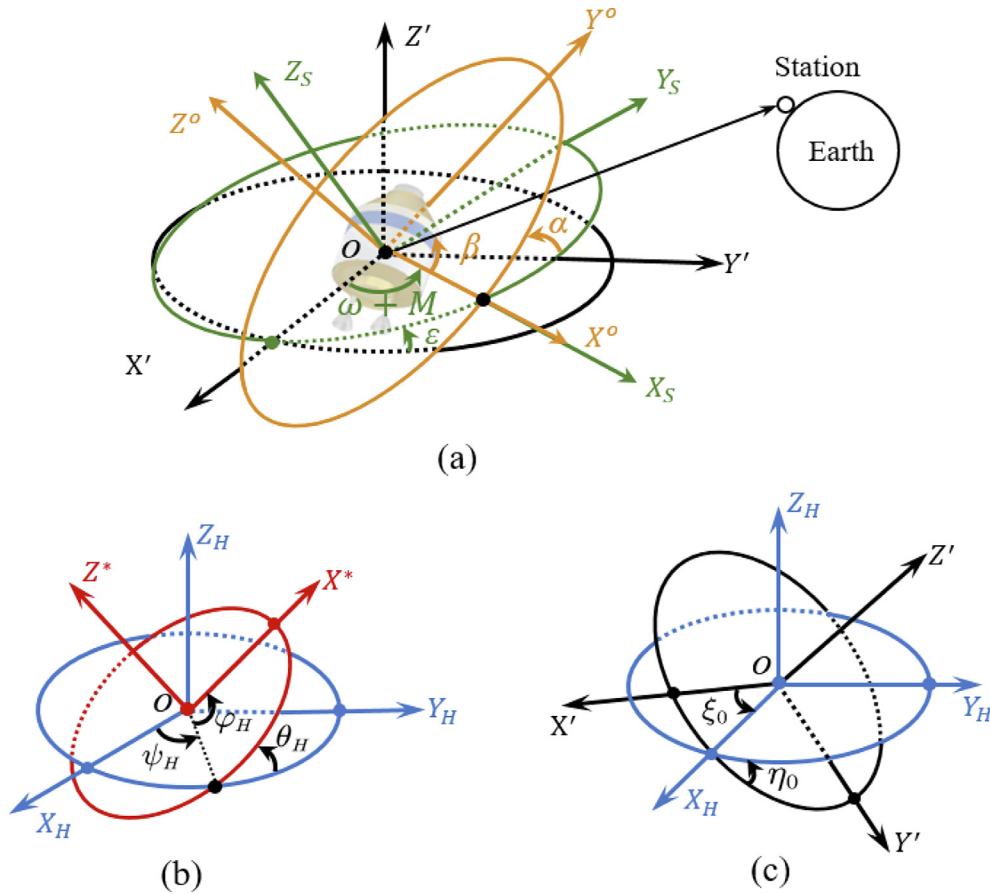


Fig. 3. Relationship of Space Coordinate Systems. (a) Schematic diagram showing the transformation from the center-of-mass-to-light-source coordinate system $oX^oY^oZ^o$ to the ecliptic coordinate system $oX_SY_SZ_S$, and then to the J2000.0 epoch equatorial coordinate system $oX'Y'Z'$. $oX^oY^oZ^o$ is the center-of-mass-to-light-source coordinate system, consistent with that in the laboratory. (b) Schematic diagram illustrating the transformation from the angular momentum coordinate system $oX_HY_HZ_H$ to the body-fixed coordinate system $oX^*Y^*Z^*$. Here, $oX_HY_HZ_H$ is the angular momentum coordinate system established with the angular momentum direction as the oZ_H axis. (c) Schematic diagram showing the transformation from the J2000.0 epoch equatorial coordinate system $oX'Y'Z'$ to the angular momentum coordinate system $oX_HY_HZ_H$.

sents the solar perihelion angular distance and M represents the mean solar perihelion angle. Vectors emanating from the center of mass toward the station and the sun are represented by n_0 and n_s , respectively. The phase angle β , is defined as the angle subtended by n_0 and n_s .

$$\beta = \arccos \frac{n_s \cdot n_0}{|n_s| \cdot |n_0|}, \quad (3)$$

In the laboratory setup, it is defined by the two vectors, one from the target to the camera stand, and the other from the target to the light source in the laboratory, as demonstrated in Fig. 1.

Within an observation arc, due to the short accumulation time, effects such as the gravity-gradient force and solar radiation pressure on the space debris are ignorable. Therefore, its angular momentum can be considered as constant. We establish the coordinate system $oX_H Y_H Z_H$ with the center of mass o as the origin and the direction of angular momentum as the oZ_H axis. The transformation matrix from the $oX_H Y_H Z_H$ coordinate system to the body-fixed coordinate system $oX^* Y^* Z^*$ is $R_z(\varphi_H)R_x(\theta_H)R_z(\psi_H)$, as depicted in Fig. 3 (b).

The transformation from the $oX' Y' Z'$ coordinate system to the $oX_H Y_H Z_H$ coordinate system is depicted in Fig. 3 (c) and the matrix can be expressed by $R_x(\eta_0)R_z(\xi_0)$. Since the angular momentum remains constant in space, ξ_0 and η_0 are constants. Hence, the transformation matrix for the point coordinates from the center-of-mass-to-light-source coordinate system $oX^o Y^o Z^o$ to the body-fixed coordinate system $oX^* Y^* Z^*$ in space is $R_z(\varphi_H)R_x(\theta_H)R_z(\psi_H)R_x(\eta_0)R_z(\xi_0)R_x(-\varepsilon)R_z(-(\omega + M))R_x(-\alpha)$.

By combining Eq. (1), we derive the relationship between the target attitude angles in the laboratory and those in space as

$$\begin{aligned} R_x(\varphi')R_y(\theta')R_z(\psi')R_y(-90)R_z(-90) &= R_z(\varphi_H)R_x(\theta_H)R_z(\psi_H) \\ R_x(\eta_0)R_z(\xi_0)R_x(-\varepsilon)R_z(-(\omega + M))R_x(-\alpha). \end{aligned} \quad (4)$$

The correlation between the target's attitude angles in space and the corresponding angles within the photometric database is systematically established as

$$\begin{aligned} R_z(\varphi)R_x(\theta)R_z(\psi) &= R_z(\varphi_H)R_x(\theta_H)R_z(\psi_H)R_x(\eta_0) \\ R_z(\xi_0)R_x(-\varepsilon)R_z(-(\omega + M))R_x(-\alpha). \end{aligned} \quad (5)$$

2.2. Photometry dataset based on laboratory test

2.2.1. The impact of visual angle on photometry

The bounding box dimensions of the model that test in the laboratory are roughly $0.3 \times 0.3 \times 0.6 \text{ m}^3$. Positioned approximately 5 m away, the camera captures the target with a visual angle of about 6.87° ($2 \times \arctan(\frac{0.6}{2 \times 5}) \approx 6.87^\circ$), depicted in Fig. 4. Assuming a telescope observes the object 2022-027B, and within the observation arc segment described in Section 3, the average slop distance is 2000 km, and the maximum dimension of the target is approximately 5.4 m, the approximate visual

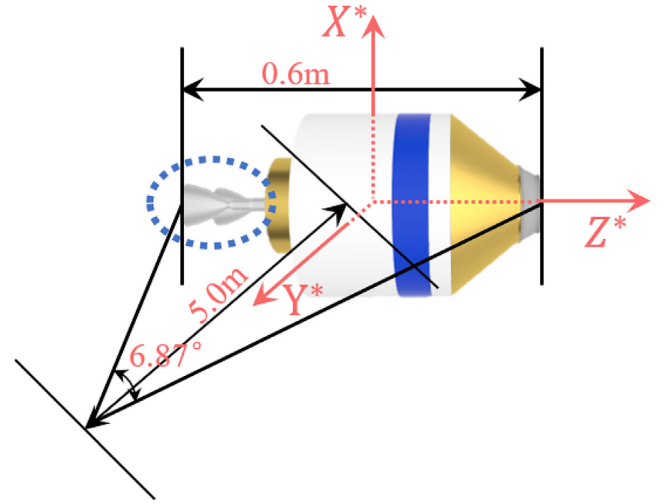


Fig. 4. Schematic diagram depicting the visual angle of the laboratory test. The section enclosed within the blue dashed line represents the tail adapter portion.

angle would be 0.557 . To acquire photometry from laboratory images, our approach involves selecting a specified target area within the captured image and then calculating the target's photometry by summing the gray value of each pixel within that area, after subtracting the gray value of the background pixels. Our previous work (Wang et al., 2023) involved a comparative analysis of this method against the 3σ algorithm and the Otsu algorithm. The findings affirmed the simplicity and high reliability of the direct subtraction method, leading us to adopt it consistently for extracting photometry from laboratory-captured images.

In that work (Wang et al., 2023), we validated the consistency between the photometry from the Povray-based numerical simulations and laboratory simulation. For the numerical simulation, we fitted the reflective properties of the actual materials and eliminated the default gamma correction setting. Regarding the laboratory simulation, we refined the photometry extraction method for the images captured within the laboratory. This comprehensive approach ensured that the numerical simulations are in agreement with the photometric data obtained from laboratory measurements. In this current endeavor, we capitalize on the ability of numerical simulations to calculate photometry at varying distances. We calculate the photometry with the target rotating in the center-of-mass-to-light-source coordinate system $oX^o Y^o Z^o$ through the matrix $R_x(\theta)R_z(\psi)$ at a phase angle of 10° . We then compare the photometry difference under two distinct visual angles, as shown in Fig. 5. The graphic demonstrates that the photometric data differences between the two visual angles diminish as θ approaches 90° , while the differences increase as θ deviates from 90° . The light curves at $\psi = 0 - 360^\circ$, $\theta = 90^\circ$ and 150° are exhibited in Fig. 6. The maximum data difference between the two visual angles is less than 0.05 magnitudes, which is significantly smaller than that caused by the target attitude change.

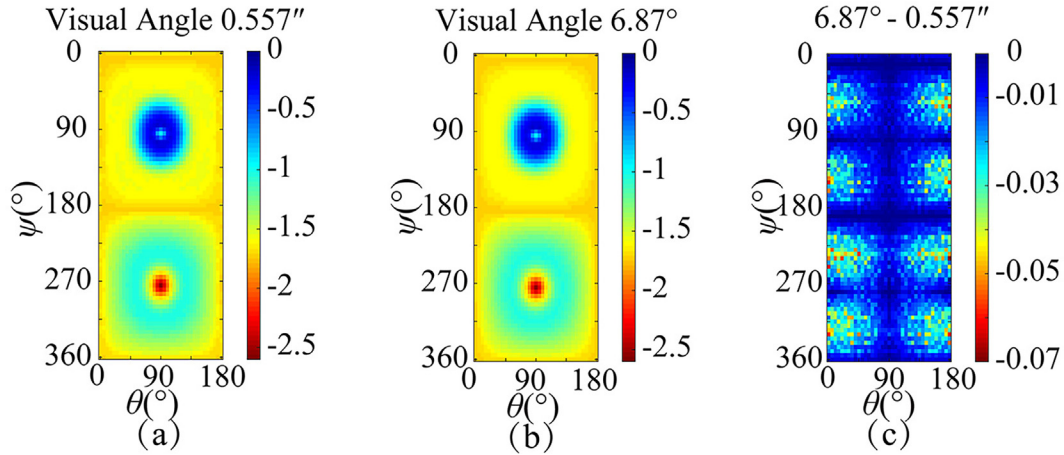


Fig. 5. Photometry data with visual angle of (a) $0.557''$, and (b) 6.87° and (c) their differences at a phase angle of 10° . Each pixel's color denotes a specific value. The target rotates within the center-of-mass-light source coordinate system $oX^oY^oZ^o$ with the rotation matrix $R_x(\theta)R_z(\psi)$. The vertical axis represents the ψ angle, while the horizontal axis depicts the θ angle.

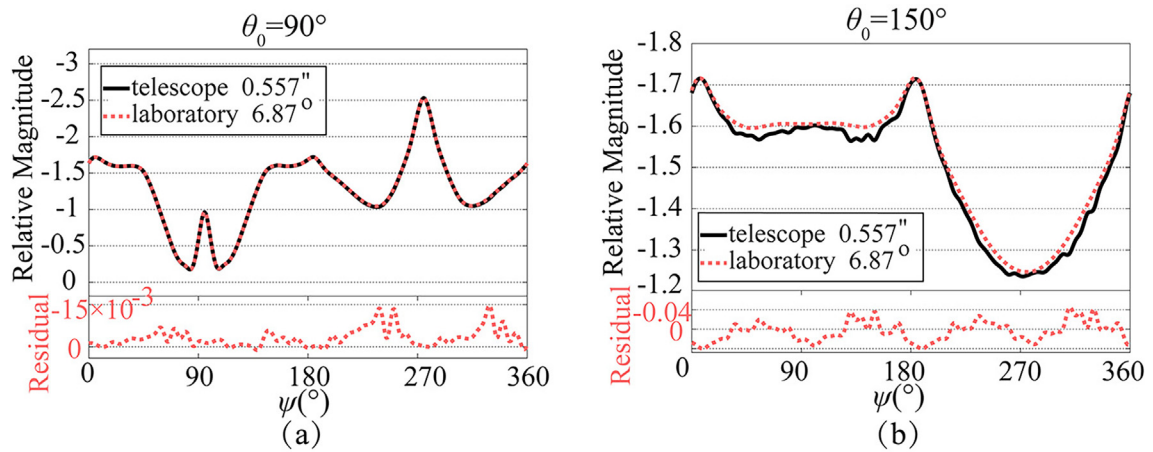


Fig. 6. Light curves for visual angles of $0.557''$ and 6.87° , with varying ψ and (a) $\theta = 90^\circ$ and (b) $\theta = 150^\circ$.

Therefore, the influence of visual angle differences between laboratory tests and telescope observations on photometry is negligible.

The primary distinctions between the telescope observations and laboratory tests are the visual angle and the uncertainty of the observation environment and equipment condition, including material aging, atmospheric turbulence fluctuations, sky-light background interference. In this paper, we categorize these latter factors as observation noise. Given the negligible impact of visual angle difference between telescope observations and laboratory tests on photometry, we can construct the photometry dataset of targets based on laboratory test to simulate that in space.

2.2.2. Constructing the photometric dataset

The target photometry is associated with four angles, including three-dimensional attitude angles (ψ, θ, φ) in the center-of-mass-to-light-source coordinate system $oX^oY^oZ^o$ and one-dimensional observation phase angle β .

Therefore, the preliminarily designed photometry dataset has four dimensions.

Now, we exam the impact of the tail adapter (the part enclosed by the blue dashed lines in Fig. 4) on photometry. The adapter is mounted at the rear end of the upper stage of the rocket (adjacent to the thrust section), and it serves multiple roles including providing a payload interface, transferring loads, facilitating payload separation, and enabling electrical integration and thermal management. Compared to the main body of the upper stage, the adapter is significantly smaller in both volume and mass. The light curve of the target rotating 90° around the Z^* axis at a phase angle of 10° tested in the laboratory, is displayed in Fig. 7 (a). The photometric variation caused by the tail adapter projection on the detector is less than 0.03 magnitudes, which is a minor figure compared to the overall photometry of -1.64 magnitudes. Conversely, we test the light curve of the target rotating 360° around the Y^* axis at an observation phase angle of 10° in the laboratory, and calculate the light curves for the complete target and the target

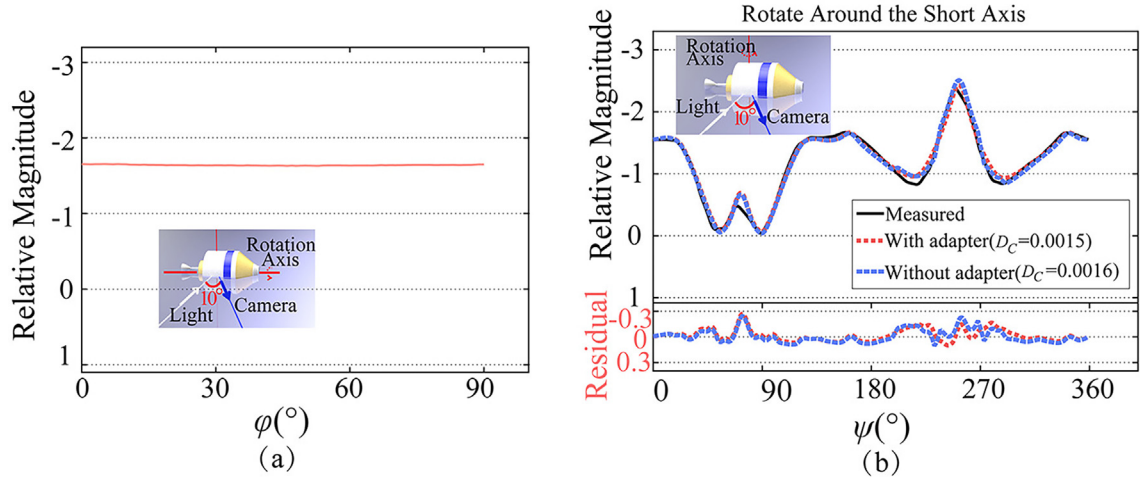


Fig. 7. Light curve of the target at a phase angle of 10°. (a) Light curve of the target when rotating 90° around Z^* axis of the main body. (b) Experimental and calculated light curves of the target when rotating 360° around the depicted Y^* axis, alongside the light curve calculated by omitting the adapter.

without the adapter under this rotation mode (Fig. 7 (b)). We discover that the photometry change caused by the adapter is less than 0.14 magnitudes, which is significantly smaller than the roughly 2.5 magnitudes variation caused by the target's main body. In Fig. 7 (b), the differences between the numerically calculated light curves and the measured one in laboratory under two scenarios are described using the cosine distance (D_C). The cosine distance between the two data sequences A and B is expressed as Eq. (6). A_i, B_i are the specific values in the sequences A, B, respectively. S_C is the cosine similarity. The shorter the cosine distance between the two data sequences, the higher the data similarity.

$$D_C = 1 - S_C = 1 - \frac{AB}{\|A\|\|B\|} = 1 - \frac{\sum_i A_i B_i}{\sqrt{\sum_i A_i^2} \sqrt{\sum_i B_i^2}} \quad (6)$$

As can be seen from Fig. 7, the impact of the adapter on photometric data usage and analysis can be disregarded. We can treat the model as a rotationally symmetric model, allowing us to eliminate the φ dimension of the photometry dataset, reducing the photometry dataset to three dimensions (ψ, θ, β) .

In theory, the photometry of (ψ, θ, φ) with any arbitrary φ can be included as the photometric value in the dataset. However, due to the occlusion of the laboratory equipment, there are inconsistencies of (ψ, θ, φ) photometry with different φ . Thus, in the dataset construction, it is essential to analyze the interference of φ angles within the laboratory environment. Through angle transformation and data processing, our goal is to make the final dataset independent of the φ angle.

We fully leverage the symmetry of the model to resolve the incomplete and unstable issues of the photometry dataset. If we directly take Eq. (1) to compute $(\psi', \theta', \varphi')$ from (ψ, θ, φ) and then remark the photometry of $(\psi', \theta', \varphi')$ into

the photometry dataset, θ' in some attitudes can exceed the laboratory test range limitation of 0–90°. For example, when $\varphi = 190^\circ$, the solved $\theta' = \arccos(\sin \varphi \sin \theta) > 90^\circ$. As a result, the photometry cannot be directly obtained by rotating the target, leading to an incomplete photometry dataset. In this case, we utilize the rotational symmetry of the model, transforming (ψ, θ, φ) into symmetrically equivalent angles $(\psi + 180^\circ, -\theta, \varphi)$. Then, $\theta' = \arccos(\sin \varphi \sin(-\theta)) > 90^\circ$, which falls within the measurable range in the laboratory.

Furthermore, at certain angles, the bracket forms a shadow area on the target (Fig. 8 (a)), or the projection of the bracket and the target overlap on the image (Fig. 8 (b)). These result in incorrect photometry extraction, ultimately leading to discontinuity in the photometry dataset. By leveraging the symmetry of the model, the photometry for (ψ, θ) is calculated in three steps. Firstly, we obtain the laboratory angles $(\psi', \theta', \varphi')$ corresponding to the attitude angles (ψ, θ, φ) . To enhance the stability of the photometry dataset in the dataset, φ varies from 0° to 355° in steps of 5°. We compute the sequences of $(\psi', \theta', \varphi')$ corresponding to each (ψ, θ, φ) . Secondly, sequences of $(\psi', \theta', \varphi')$ with unreliable photometry are excluded. We assess whether the bracket lies along the light path from the light source to the target, based on the dimensions and angles of the target and the rotation axis bracket. If so, the shadow of the bracket obstructs the target, thus the corresponding angles need to be excluded (see Fig. 8 (a)). We determine whether the target, the pitch-axis support, and the camera are collinear, based on the position and angle of the camera and the position of the target. If they are collinear, the bracket and the target have overlapping images on the camera, thus the corresponding angles need to be excluded, either (see Fig. 8 (b)). Thirdly, we compute the average photometry of the remaining sequences of $(\psi', \theta', \varphi')$ after screening. This average serves as the value for (ψ, θ) in the photometry dataset.

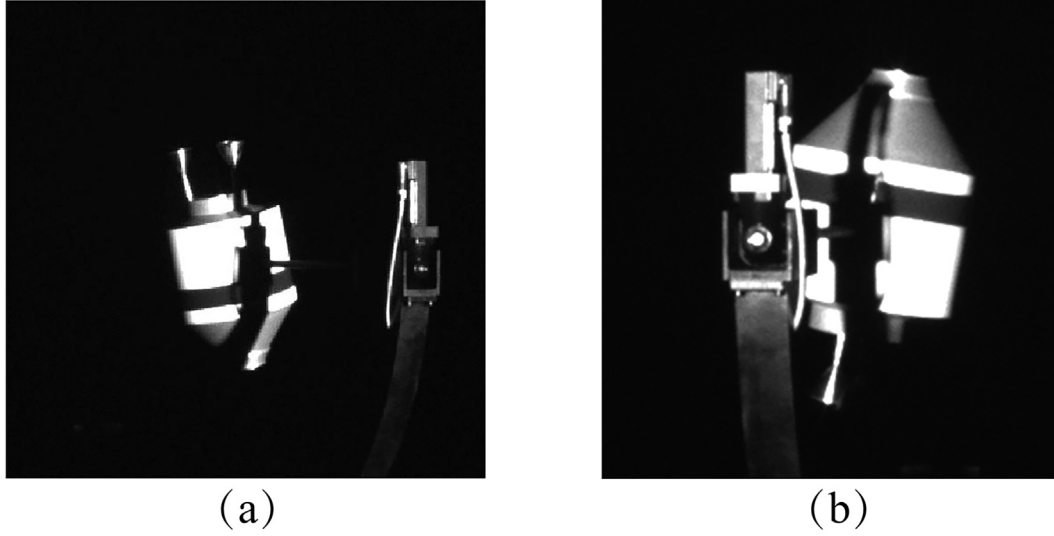


Fig. 8. Illustrations of how the bracket can affect target information extraction during laboratory testing. (a) An instance where the bracket obscures the target. (b) An instance where the bracket and target imaging overlap on the camera.

Based on the photometry tested in the laboratory and the angle conversion relationship, the final photometry dataset sample constructed is shown in Fig. 9. The magnitudes value in Fig. 9, Fig. 5, and Fig. 7 are normalized to the faintest signal obtained during laboratory testing, meaning the highest magnitude in the database is set at 0 magnitude.

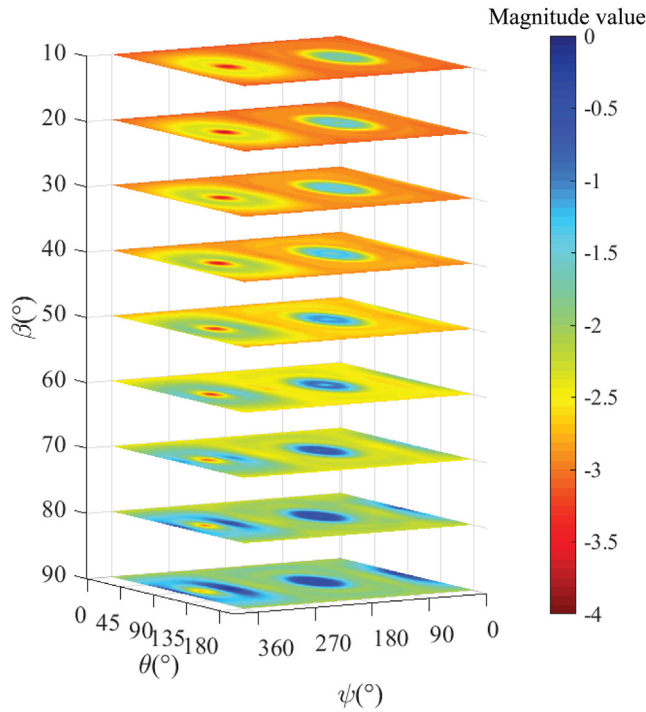


Fig. 9. Profile map of photometry dataset. The coordinates represent the attitude angles ψ and θ of the body-fixed coordinate system in the centroid-light source coordinate system, as well as the observation phase angle β . The color of each pixel represents the magnitude of photometry at the corresponding angle.

2.3. Analysis of attitude angle variation rules

We assume that the material density distribution of the CZ-4C upper stage is uniform. To simplify the analysis of attitude angles rules and to ensure consistency with the photometry database, the adapter is disregarded. The principal moments of inertia are approximated as $I_{x^*} : I_{y^*} : I_{z^*} = A : B : C = 4 : 4 : 3$, with the oZ^* axis as the dynamic symmetry axis of the target. The three components of angular momentum can be individually expressed as

$$H_{x^*} = Ap, H_{y^*} = Bq, H_{z^*} = Cr, \quad (7)$$

and the magnitude of the angular momentum is

$$H = \sqrt{H_{x^*}^2 + H_{y^*}^2 + H_{z^*}^2}, \quad (8)$$

where p, q and r represent the projections of the angular velocities on the body-fixed coordinate axes oX^*, oY^* and oZ^* , respectively. For a dynamically symmetrical rigid body ($A = B$), the projection of the target angular velocity on its axis of dynamic symmetry is constant

$$r = r_0, \quad (9)$$

and the following relationship holds

$$\begin{cases} Cr = H \cos \theta_H \\ \theta_H = \theta_{H0} \\ \dot{\psi}_H \equiv \frac{H}{A} \equiv \frac{Cr_0}{A \cos \theta_{H0}} \\ \dot{\varphi}_H \equiv \frac{(A-C)r_0}{A} \\ \psi_H = \psi_{H0} + \dot{\psi}_H t \\ \varphi_H = \varphi_{H0} + \dot{\varphi}_H t \end{cases} \quad (10)$$

3. Attitude inversion based on photometry

As indicated by Eq. (5), the attitude parameters of the target in space are denoted as $(\alpha, \omega, M, \varepsilon, \xi_0, \eta_0, \psi_H, \theta_H, \varphi_H)$. For a given observation arc, $(\alpha, \omega, M, \varepsilon)$ and phase angle β can be calculated from observation time, station location, and target orbital elements. Combining these with Eq. (10), we can determine the sequence of (ψ, θ, β) uniquely by introducing $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$. Therefore, the attitude inversion is to search for $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ that best matches the simulated light curve within the photometry dataset, under the constraint of the rotation dynamics rules. We must take into account a particular scenario, where $\xi_0^\# = 180^\circ + \xi_0, \eta_0^\# = 180^\circ - \eta_0, \psi_{H0}^\# = 360^\circ - \psi_{H0}, \theta_{H0}^\# = 180^\circ - \theta_{H0}$. The attitude parameters $(r_0, \xi_0^\#, \eta_0^\#, \psi_{H0}^\#, \theta_{H0}^\#)$ and $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ correspond to identical sequences of $(\psi^\#, \theta^\#)$ and (ψ, θ) . Therefore, to eliminate such equivalent solutions, we constraint η_0 within the range of $0 - 90^\circ$.

We simulate observations of 2022-027B from Yao'an Observatory (101.18 E 25.53 N 2000 M), with the start times of the three arcs being 22:55:12 UTC on January 3, 2023, 22:22:42 UTC on January 5, 2023 and 23:02:15 UTC on January 6, 2023. The attitude parameters for the start of the first arc are set as $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0}) = (2.6667^\circ/\text{s}, 30.00^\circ, 40.00^\circ, 20.00^\circ, 60.00^\circ)$. The photometry sequence in the photometry dataset is obtained through Eq. (10) and Eq. (5). To account for uncertainties in telescope observations, noise is added to photometry. Since the laboratory-tested light curves have not been calibrated with the telescope-observed ones, there is a certain discrepancy between them. However, this does not affect the analysis of attitude parameters based on relative photometric changes. The light curves of the three arcs are shown in Fig. 10.

We employ genetic algorithm to search for the attitude parameters of the simulated light curves. Assuming P_s is the simulated observed photometry and P_i is the inversed

photometry in the photometry database, the fitness function for the genetic algorithm in this study is

$$\text{Fitness} = \frac{1}{2} \left(\frac{\sum_{k=1}^N \frac{|P_{sn_k}|}{(P_{sn_k} - P_{in_k})^2 + 0.01}}{\sum_{k=1}^N |P_{sn_k}|} + \frac{1}{0.01 + 0.01 \left(\frac{\sum_{k=1}^N |P_{sn_k} - P_{in_k}|}{N} \right)} \right), \quad (11)$$

where $P_{sn} = P_s - \bar{P}_s$ and $P_{in} = P_i - \bar{P}_i$. The maximum of this fitness function is 100, indicating that P_s and P_i only have a translational relationship, with the magnitude and trend of photometric changes being completely consistent.

We search the attitude parameters for both the light curve without noise (the black line in Fig. 10) and the light curve with 0.5 magnitude noise (the red line in Fig. 10). Subsequently, we filter out the top 5% of attitude parameters based on their fitness ranking. As evident in Fig. 11, a strong correlation exists between r_0 and ψ_{H0} . The individual solutions for r_0 and ψ_{H0} obtained from the three arcs each cover a wide range. However, the joint solution of the three arcs constrains them within a narrower range, as shown by the purple area in Fig. 11. The solved ranges of attitude parameters are presented in Table 1. Secondary grids are harnessed to get more accurate attitude parameters via the genetic algorithm. For data with 0.5 magnitude noise, the parameters spanning across wide region are due to the lower fitness threshold for screening. However, the final solution resides in the center region, which is close to the preset value. Hence, a judicious expansion of the parameter ranges does not compromise the accuracy of the final outcome.

We will utilize this technology to perform attitude inversion calculations on simulated light curves with varying levels of noise. For each light curve, 600 iterations will be

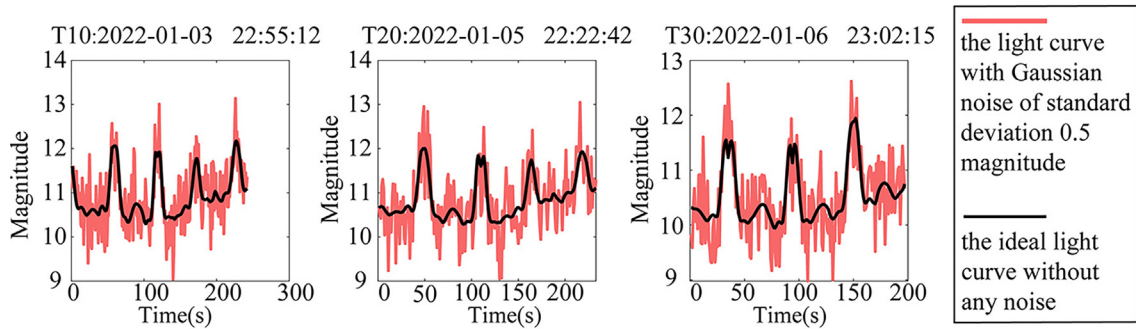


Fig. 10. Light curves of the simulated observation. The black line in the figure signifies the photometry without observation noise, whereas the red line indicates the photometry with an added noise of 0.5 magnitudes.

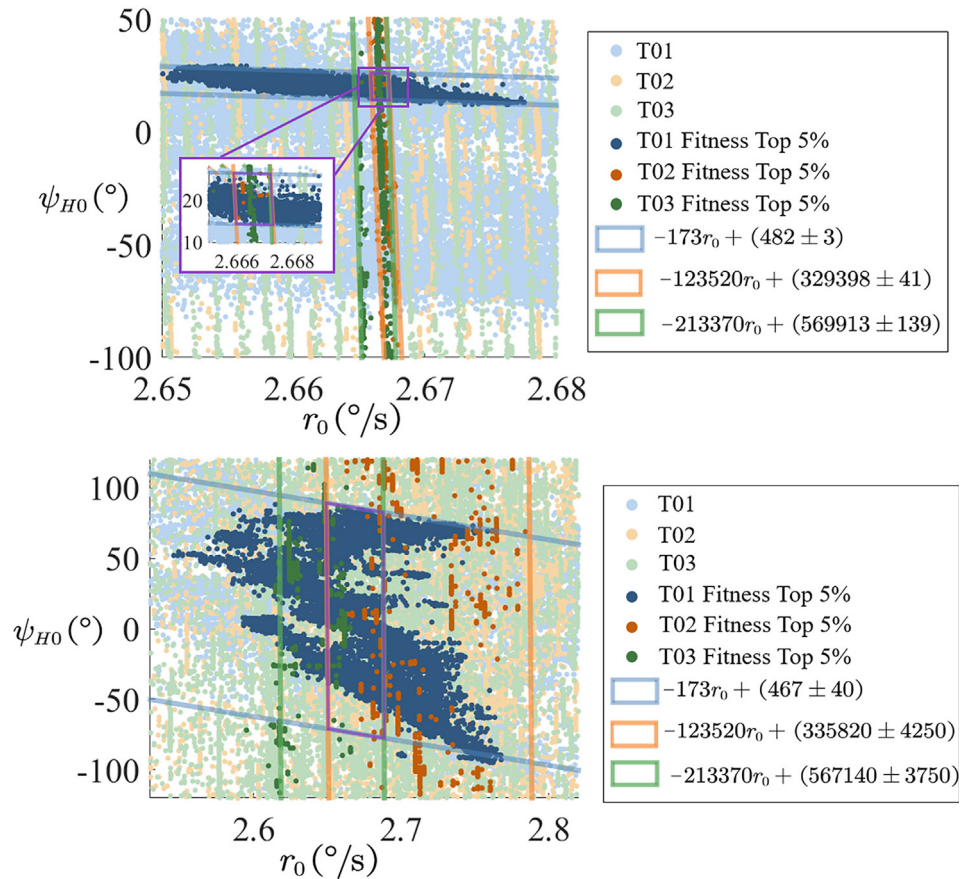


Fig. 11. The preliminary calculation of r_0 and ψ_{H0} distribution. The top presents the results from light curves without observation noise, while the bottom displays the results from light curves with a 0.5 magnitude noise. The light blue, light yellow, and light green points represent all the results from arc T_{01} , T_{02} , and T_{03} , respectively. The dark blue, dark yellow, and dark green points correspond to the top 5% results in terms of fitness from arcs. The blue, yellow, and green framed areas indicate the envelope areas of the corresponding dark points. The overlapping area of these framed areas is shown in purple.

conducted to obtain a preliminary range of parameters, followed by 200 calculations with a secondary network to acquire the refined attitude parameters. We will investigate the algorithm's resistance to noise, the sensitivity of the light curves to different parameters, and enhance the accuracy of results by preprocessing the light curve data.

4. Discussion and improvements of the inversion algorithm

4.1. Noise resistance of the inversion algorithm

To assess the noise resilience of our algorithm, we invert the light curves across a range of noise levels, from 0.0 to 1.0 magnitudes, without applying any preprocessing tech-

niques such as smoothing or filtering to the photometric data. Given that the noise inherent in the simulated data influences the fitness evaluation, we aim to compare the quality of inversion results under different noise levels based on a uniform standard. Therefore, we compute the fitness of the light curve associated with the inversed attitude parameters with respect to the ideal simulated photometry (the black line in Fig. 10), as shown in Table 2. The deviations between the attitude parameters obtained from the inversion under different noise levels and the pre-set attitude values are shown in Table 3.

The inversed attitude parameters are shown as red lines in Fig. 12, and the recalculated fitness are indicated by the green hollow circles in Fig. 12. The results indicate that as

Table 1
Range of inverted attitude parameter and final calculation results obtained from inversion algorithm.

Parameter	Range of Parameter		Final Results	
	Noise = 0.0 Magnitude	Noise = 0.5 Magnitude	Noise = 0.0 Magnitude	Noise = 0.5 Magnitude
r_0 (°/s)	2.6666 ± 0.001	2.6688 ± 0.0196	2.6667	2.6556
ξ_0 (°)	29.68 ± 7.44	34.59 ± 8.98	30.02	28.95
η_0 (°)	40.96 ± 3.78	33.99 ± 12.59	39.98	39.84
ψ_{H0} (°)	22.90 ± 5.55	5.84 ± 83.40	19.98	23.72
θ_{H0} (°)	59.41 ± 3.90	57.44 ± 9.90	60.03	55.38

Table 2
Direct inversion results.

Observation Noise(Magnitude)	r_0 (°/s)	ξ_0 (°)	η_0 (°)	ψ_{H0} (°)	θ_{H0} (°)	Fitness
Preset Value	2.6667	30.00	40.00	20.00	60.00	100.00
0.0	2.6667	30.03	39.98	19.98	60.03	99.98
0.1	2.6667	31.14	40.35	18.69	59.59	98.54
0.2	2.6667	29.41	40.15	19.50	59.46	97.33
0.3	2.6666	33.83	43.18	17.65	58.32	85.09
0.4	2.6666	31.03	45.76	25.79	55.33	81.98
0.5	2.6556	28.95	39.84	23.72	55.38	77.19
0.6	2.6611	35.83	31.94	10.00	56.60	71.34
0.7	2.6612	26.03	41.94	21.49	55.63	81.90
0.8	2.6666	36.19	40.74	13.69	55.21	78.98
0.9	2.7479	27.49	33.94	10.00	54.34	67.14
1.0	2.7479	32.08	39.51	21.24	55.08	77.01

Table 3
The deviation between the direct inversion results and the preset values.

Observation Noise(Magnitude)	$ \Delta r_0 $ (°/s)	$ \Delta \xi_0 $ (°)	$ \Delta \eta_0 $ (°)	$ \Delta \psi_{H0} $ (°)	$ \Delta \theta_{H0} $ (°)
0.0	0.0000	0.03	0.02	0.02	0.03
0.1	0.0000	1.14	0.35	1.31	0.41
0.2	0.0000	0.59	0.15	0.50	0.54
0.3	0.0001	3.83	3.18	2.35	1.68
0.4	0.0001	1.03	5.76	5.79	4.67
0.5	0.0110	1.05	0.16	3.72	4.62
0.6	0.0056	5.83	8.06	10.00	3.40
0.7	0.0055	3.97	1.94	1.49	4.37
0.8	0.0001	6.19	0.74	6.31	4.79
0.9	0.0812	2.51	6.06	10.00	5.66
1.0	0.0055	2.08	0.49	1.24	4.92

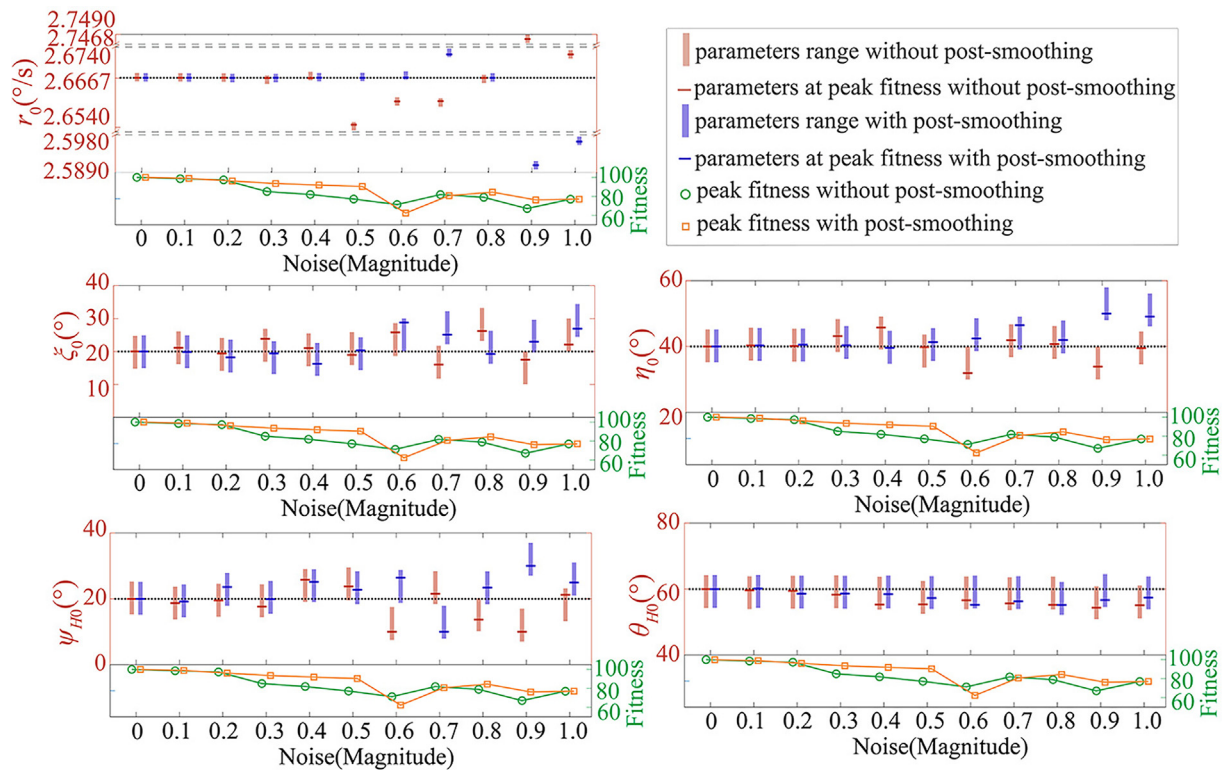


Fig. 12. Analysis of inversion results with direct inversion and post-smoothing inversion under various noise levels. In the upper section of each figure, the black dashed line denotes the preset values. The red and blue short lines correspond to the inversed attitude parameters corresponding to the inversions without and with post-smoothing, respectively. The rectangles indicate the intervals of attitude parameter variations at the peak fitness, ranging within (0.0020°/s, 10°, 10°, 10°, 10°). In the lower section of each figure, the green hollow circles and yellow hollow squares represent the fitness of the inversions without and with post-smoothing, respectively.

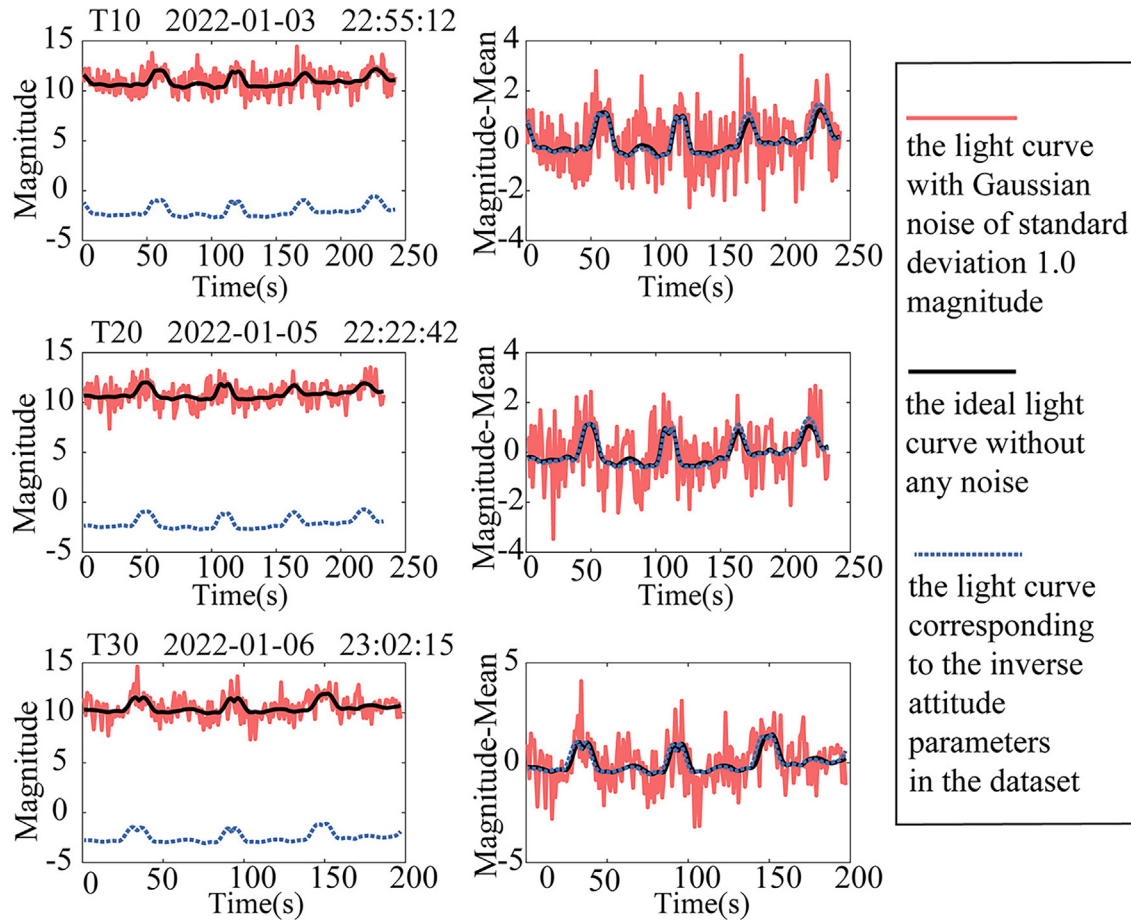


Fig. 13. Comparison of observed photometry with 1.0 magnitude noise and inverted photometry. The graph on the left portrays the raw photometric data (uncorrected), whereas its counterpart on the right depicts the same data following a mean correction. Each row of the triptych corresponds to a distinct observational time.

the observation noise varies from 0.0 to 1.0 magnitude, the maximum deviations of the attitude parameters ($r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0}$) from the preset values are $|\Delta r_0|_{\max} = 0.0812^\circ/\text{s}$, $|\Delta \xi_0|_{\max} = 6.19^\circ$, $|\Delta \eta_0|_{\max} = 8.06^\circ$, $|\Delta \psi_{H0}|_{\max} = 10.00^\circ$ and $|\Delta \theta_{H0}|_{\max} = 5.66^\circ$. As the observational noise increases from 0.0 to 0.6 magnitudes, the fitness of the inverted attitude results gradually decreases. When the observational noise increases from 0.6 to 1.0 magnitudes, the fitness fluctuates within a certain range. Taking the light curve with maximum noise level of 1.0 magnitude as

an example, Fig. 13 shows the simulated observed light curve with noise, the ideal observed light curve without noise, and the light curve associate with the inversed attitude parameters in the photometry dataset. For the simulated light curve with a variation amplitude of approximately 2.5 magnitudes, even when the noise reaches 1.0 magnitudes, the inversed light curve still exhibits high conformity with the original simulated ideal light curve without noise. This suggests a robust noise resistance of the proposed inversion algorithm.

Table 4
Change in fitness caused by changes in attitude parameters.

Observation Noise(Magnitude)	r_0 (0.0020°/s)	ξ_0 (10.00°)	η_0 (10.00°)	ψ_{H0} (10.00°)	θ_{H0} (10.00°)
0.0	36.29	4.59	11.98	2.59	25.26
0.1	25.03	0.99	2.73	0.43	12.08
0.2	16.50	0.36	1.21	0.22	6.31
0.3	12.73	0.19	0.31	0.10	3.77
0.4	8.59	0.20	0.40	0.06	2.47
0.5	6.65	0.08	0.19	0.12	1.92
0.6	5.55	0.01	0.16	0.13	2.25
0.7	4.07	0.09	0.23	0.03	1.34
0.8	3.67	0.06	0.19	0.04	1.67
0.9	2.80	0.11	0.13	0.05	0.92
1.0	2.62	0.06	0.07	0.02	0.74

4.2. Sensitivity of fitness to attitude parameters

To evaluate the sensitivity of fitness to different attitude parameters, we take a range of $(0.0020^\circ/\text{s}, 10.00^\circ, 10.00^\circ, 10.00^\circ, 10.00^\circ)$ for the parameters $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ at the positions corresponding to the maximum fitness. This enables an analysis of the influence exerted by attitude parameters on fitness. Variations in the fitness are detailed in Table 4. Fig. 14 illustrates the relationship

between the attitude parameters and the fitness under conditions of no observation noise and observation noise of 0.5 magnitude. For easier visualization, Fig. 15 exhibits the changes in fitness resulting from a 10° range in each four angle parameters. It is observed that as noise escalates, the sensitivity of fitness to parameters diminishes. Moreover, the fitness is more sensitive to θ_{H0} and η_0 compared to ξ_0 and ψ_{H0} . As a result, when the four angle parameters $(\xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ cannot be entirely resolved,

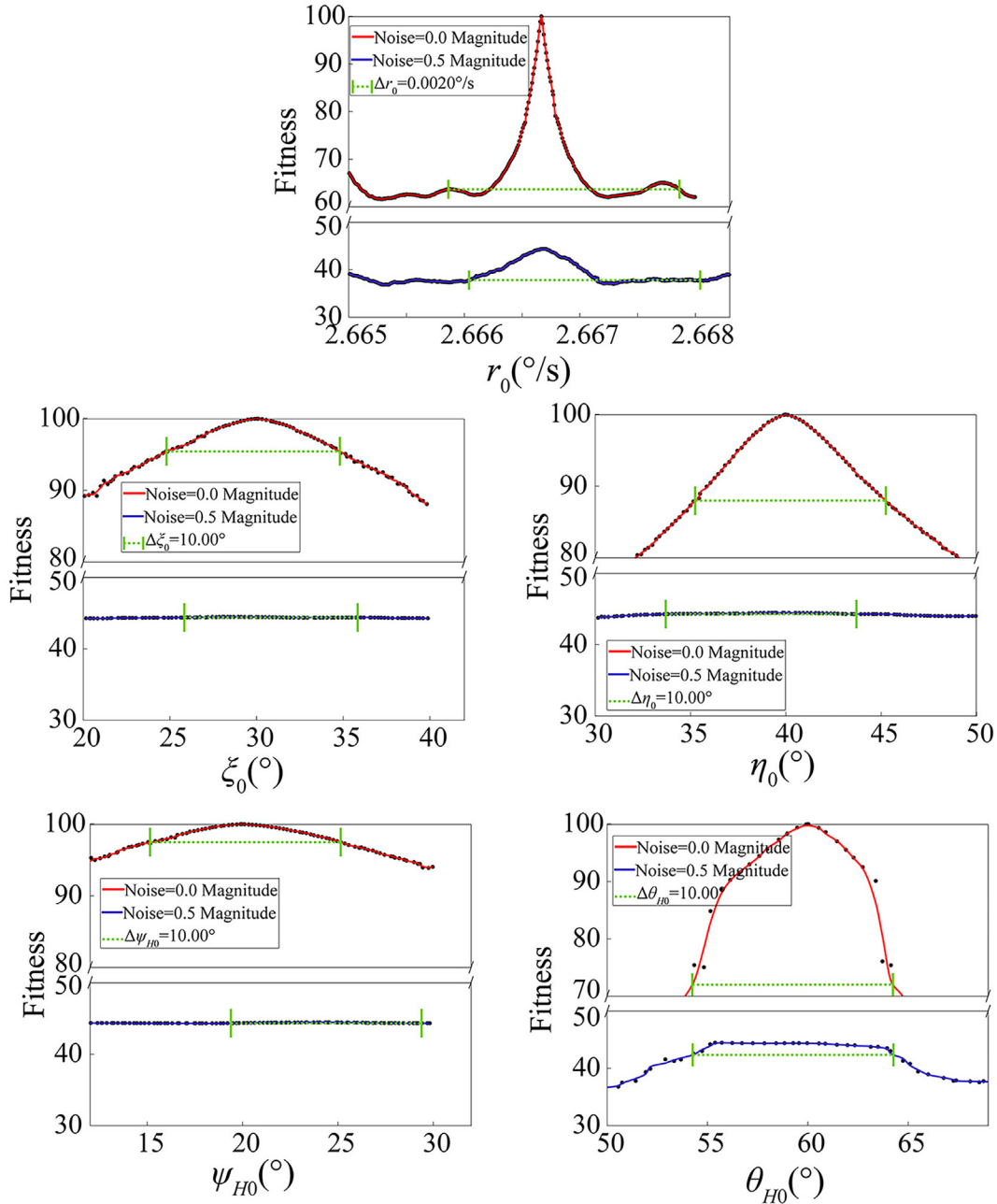


Fig. 14. The relationship between attitude parameters values and fitness under conditions of no observation noise and 0.5 magnitude observation noise. The red line represents the solution results without observation noise. The blue line illustrates the solution results with an observation noise of standard deviation 0.5 magnitude. The green line indicates the intervals of attitude parameters $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ at the peak fitness position, ranging within $(0.0020^\circ/\text{s}, 10.00^\circ, 10.00^\circ, 10.00^\circ, 10.00^\circ)$. The black dots represent the attitude parameter values calculated from the inversion. These values are discrete. In order to determine the specific range of attitude parameters $(r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0})$ within $(0.0020^\circ/\text{s}, 10.00^\circ, 10.00^\circ, 10.00^\circ, 10.00^\circ)$ at the peak fitness position, we performed simple smoothing and interpolation calculations on the discrete results, resulting in the red lines and blue lines.

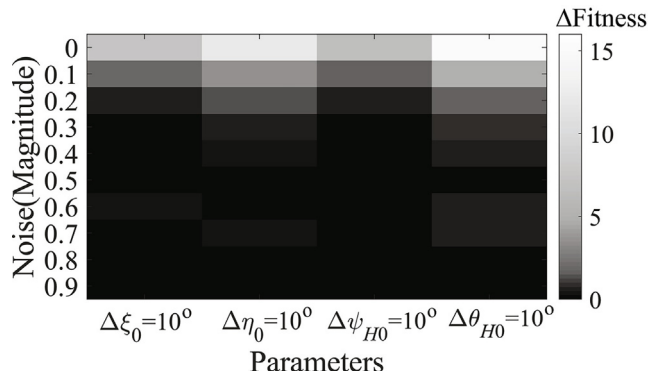


Fig. 15. Under direct inversion, the variation in fitness corresponding to a 10° range in the angles of attitude parameters near the peak fitness position.

particular emphasis should be placed on the values of θ_{H0} and η_0 .

4.3. The improved inversion algorithm with smoothing preprocessing

Within the inversion of attitude from photometry, the noise inherent in the observations is incorporated as an integral component in computation of fitness. This incorporation implies that the noise can exert an influence over

the selection of attitude parameters during the inversion. In response, we propose a preprocessing step for photometric data by smoothing prior to inversion. This step aims to mitigate some of the noise interferences, thereby enhancing the accuracy of the attitude parameters. Table 5 presents the inversed results with post-smoothing, when the noise increases from 0.0 to 1.0 magnitude. When the observation noise escalates from 0.0 to 1.0 magnitude the deviation between the inversed attitude parameters ($r_0, \xi_0, \eta_0, \psi_{H0}, \theta_{H0}$) after smoothing and the preset values are shown in Table 6, and the maximum divergences are $|\Delta r_0|_{\max} = 0.0758^\circ/\text{s}$, $|\Delta \xi_0|_{\max} = 8.83^\circ$, $|\Delta \eta_0|_{\max} = 10.00^\circ$, $|\Delta \psi_{H0}|_{\max} = 10.00^\circ$ and $|\Delta \theta_{H0}|_{\max} = 4.83^\circ$. Parallel to the analysis of inversion without post-smoothing of photometry, we compute the fitness of the light curve associated with the inversed attitude parameters with respect to the simulated photometry without noise (the black line in Fig. 10).

The inversed attitude parameters and newly computed fitness from both the inversions without and with post-smoothing are illustrated in Fig. 12. Apart from the light curve without noise, which undergoes a signal loss from post-smoothing leading to a decrease in attitude accuracy, the attitude accuracy and fitness value with smoothing are superior to those without smoothing for the majority of light curves involving noise. Therefore, conducting attitude

Table 5
Results of attitude version after data smoothing.

Observation Noise(Magnitude)	r_0 ($^\circ/\text{s}$)	ξ_0 ($^\circ$)	η_0 ($^\circ$)	ψ_{H0} ($^\circ$)	θ_{H0} ($^\circ$)	Fitness
Preset Value	2.6667	30.00	40.00	20.00	60.00	—
0.0	2.6667	30.02	40.00	19.98	60.00	100.00
0.1	2.6667	29.84	40.34	19.15	60.13	98.88
0.2	2.6667	28.24	40.53	23.60	58.57	96.14
0.3	2.6667	29.41	40.39	19.90	58.62	93.52
0.4	2.6667	26.23	39.64	25.15	58.45	91.94
0.5	2.6667	30.31	41.36	22.73	57.28	90.41
0.6	2.6667	38.83	42.47	26.43	55.21	62.42
0.7	2.6722	35.09	46.48	10.00	56.27	80.74
0.8	2.6667	29.24	42.02	23.42	55.17	84.49
0.9	2.5909	32.98	50.00	30.00	56.67	76.24
1.0	2.5964	36.92	49.11	24.93	57.38	77.06

Table 6
The deviation between the inversion results after smoothing and the preset values.

Observation Noise(Magnitude)	$ \Delta r_0 $ ($^\circ/\text{s}$)	$ \Delta \xi_0 $ ($^\circ$)	$ \Delta \eta_0 $ ($^\circ$)	$ \Delta \psi_{H0} $ ($^\circ$)	$ \Delta \theta_{H0} $ ($^\circ$)
0.0	0.0000	0.03	0.00	0.02	0.00
0.1	0.0000	0.16	0.34	0.85	0.13
0.2	0.0000	1.76	0.53	3.6	1.43
0.3	0.0000	0.59	0.39	0.10	1.38
0.4	0.0000	3.77	0.36	5.15	1.55
0.5	0.0000	0.31	1.36	2.73	2.72
0.6	0.0000	8.83	2.47	6.43	4.79
0.7	0.0055	5.09	6.48	10.00	3.73
0.8	0.0000	0.76	2.02	3.42	4.83
0.9	0.0758	2.98	10.00	10.00	3.33
1.0	0.0703	6.92	9.11	4.93	2.62

inversion from telescope-observed photometry following the preprocessing techniques such as smoothing, filtering, and other noise reduction methods, can enhance the accuracy of attitude results.

5. Conclusion

This paper explores the inversion algorithm of attitude parameters from photometry, based on the photometry tested in the laboratory. Based on the analysis of light curves, it has been verified that the influence of visual angle difference between laboratory test and telescope observation on photometry is ignorable. Drawing upon the symmetry of the model and angle conversion relationships, a succinct and reliable CZ-4C photometry dataset is constructed. We then calculate the light curve from the dataset using preset attitude parameters and involve different levels of noise to simulate observed photometry. The genetic algorithm is adopted to inverse the attitude parameters from photometry, adhering to the constraint of non-external torque. Our exploration into the variations of fitness with attitude parameters reveals a higher sensitive to η_0 and θ_{H0} in comparison to ξ_0 and ψ_{H0} . Interestingly, the algorithm exhibits robust resistance to noise, maintaining a high accuracy in its solutions, even under an observation noise of 1.0 magnitudes. Furthermore, if the photometry is smoothed before attitude inversion, the inversed attitude parameters are closer to the preset values than those obtained from the inversion without smoothing. Comparing to previous attitude analysis, the inversion algorithm in this paper requires observed data from merely three arcs. It is less demanding in terms of data volume and characteristic, and the accuracy of results has been validated against preset values. In summary, this research offers a compelling method for future attitude research on telescope observed data.

However, this study has not yet analyzed the impact of sampling factors on the results, such as the sampling frequency or the number of data points within an observation arc. Future solutions for telescope observed photometry may need to take these factors into account.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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